

Rahul Kolla

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Summary

Agentic AI Developer and Computer Science graduate student specializing in building autonomous AI agents, multi-step reasoning pipelines, and LLM-powered systems. Hands-on experience designing agentic workflows with LangChain, tool-use orchestration, RAG architectures, and local LLM deployment via Ollama. Passionate about engineering intelligent systems that act, reason, and adapt – bridging cutting-edge AI research with production-grade backend infrastructure.

Education

University of North Carolina at Charlotte
Master of Science in Computer Science

Charlotte, NC
Expected May 2026

Vellore Institute of Technology – AP
Bachelor of Technology in Computer Science
CGPA: 8.86 / 10.0

Amaravati, India
2020 – 2024

Technical Skills

Agentic AI & GenAI: Autonomous AI Agents, LangChain Agents, Tool-Use Orchestration, RAG Workflows, Multi-Step Reasoning, Prompt Engineering, OpenAI APIs, Ollama, LLMs, Hugging Face

AI & ML: Machine Learning, Deep Learning (CNN, LSTM), Computer Vision, LLM Fine-Tuning

Programming: Python, TypeScript, JavaScript, Java, SQL, Bash

Web & Backend: FastAPI, Flask, Node.js, Express, RESTful APIs, MERN Stack, Microservices, JWT Auth

DevOps & Cloud: Docker, AWS, CI/CD, Ansible, Git/GitHub, Linux Environments

QA & Testing: Robot Framework, Selenium, Manual Testing, Bug Tracking, SAP BTP / ABAP / S/4HANA

Data & Analytics: Pandas, NumPy, Matplotlib, Tableau, SQL Workbench

Experience

QA Analyst Intern

AiFA Labs – Cerebro SASA Platform

Charlotte, NC
Sep 2025 – May 2026

- Executed manual and automated test suites using Robot Framework and Selenium on an AI-driven SAP automation platform, ensuring reliability across enterprise workflows.
- Resolved client-reported bugs through structured issue management with cross-timezone teams; contributed QA coverage across SAP BTP, ABAP, and S/4HANA in Agile sprints.

Backend Developer Intern

AiFA Labs

Hyderabad, India
Jan 2024 – Jul 2024

- Built scalable backend services for the CEREBRO platform using Python, Flask, and RESTful APIs, supporting real-time AI-driven data operations.
- Automated data ingestion and logging workflows, improving pipeline response efficiency by 30%; delivered data-driven features translating complex requirements into production-ready implementations.

Associate Technical Lead – WIOS Club

VIT-AP University

Amaravati, India
Jan 2023 – Aug 2023

- Mentored students in full-stack development; oversaw MERN stack deployments and coordinated technical hackathons fostering peer-led innovation.

Projects

FERRET – Autonomous AI Research Agent

Built an end-to-end agentic AI system that autonomously searches the web, synthesizes information across multiple sources, and generates structured research reports. Features a full web UI with live progress tracking, batch query processing, multi-format export (PDF, Markdown, JSON), and error recovery – demonstrating agentic tool-use, orchestration, and real-time feedback loops.

AI-Powered Smart Document Intelligence System

Engineered a local end-to-end RAG platform using FastAPI, LangChain, Ollama, and SQLite – enabling intelligent PDF querying, semantic search, and document summarization without cloud dependencies.

Automated Security Pipeline for SIEM and IDS

Designed and automated a real-time threat detection pipeline using Ansible to orchestrate Wazuh (SIEM/XDR) and Suricata (IDS), enabling proactive monitoring across distributed infrastructure.

Email Spam Detection Using Deep Learning

Built a spam classification system using CNN and LSTM models, integrated via Flask API and deployed on Google App Scripts for production email filtering.

AI-Based Tumor Heterogeneity Detection (*Patented*)

Proposed and implemented an AI-driven approach for quantifying intra-tumor heterogeneity using image-based biomarkers – resulting in a filed patent supporting healthcare diagnostic innovation.

Certifications

Hugging Face Agents Course	<i>Hugging Face, 2025</i>
Microsoft Azure Fundamentals (AZ-900)	<i>Microsoft, 2024</i>
Deep Learning Institute	<i>NVIDIA, 2024</i>
Prompt to Prototype	<i>Google, 2025</i>
Web Development Bootcamp	<i>Udemy, 2023</i>